

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3		(i)		Substitution: $V=IR = 2 \times 2$ (1) $= 4$ [V] (1) NB $12\text{ V} \div 3 = 4\text{ V} \rightarrow 0$ marks unless justified, e.g. by a potential divider argument.	1	1		2	2	2
		(ii)		Either: voltage across parallel pair $= 12 - 4$ ecf $= 8$ [V] (1) NB. If 6 V is used [possible ecf from (i)], $12 - 6 = 6$ must be shown to get any credit Substitution: current through $R_2 = \frac{V}{R_2} = \frac{8 \text{ ecf}}{12}$ (1) $= 0.67$ [A] (1) Current through $R_3 = 2 - 0.67$ ecf $= 1.33$ [A] (1) {or $\frac{V}{R_3} = \frac{8 \text{ ecf}}{6} = 1.33$ [A] for final mark} (1) OR to get the first mark: For first mark: Resistance of parallel pair $= 4\ \Omega$ (e.g. using $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$) <u>and</u> $V_{\text{parallel}} = IR_{\text{parallel}} = 2 \times 4 = 8$ [V] (1) <i>OR complete answer as follows:</i> Statement: currents add to 2 A (1) Statement current in the ratio 2:1 or 1:2 (1) Substitution: Currents in two arms of parallel circuit $= 0.67\text{ A}$ and 1.33 A (1) Correct allocation: Current in $12\ \Omega = 0.67\text{ A}$; Current in $6\ \Omega = 1.33\text{ A}$ (1) NB. Current not adding to 2 A in ratio 2:1 $\rightarrow 0$ marks. NB. $I_2 = 0.33\text{ A}$ and $I_3 = 0.67\text{ A} \rightarrow 0$ marks [arises from use of 4 V]	1	1 1 1		4	4	4
				Question 3 total	2	4	0	6	6	6